

EXPERIMENTS

```
Database changed
mysql>
mysql> CREATE TABLE department (
    ->     dept_id INT,
    ->     dept_name VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE student (
    ->     student_id INT,
    ->     student_name VARCHAR(50),
    ->     dept_id INT,
    ->     marks INT
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> INSERT INTO department VALUES
    -> (1, 'CSE'),
    -> (2, 'ECE'),
    -> (3, 'IT'),
    -> (4, 'MECH');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO student VALUES
    -> (101, 'Arun', 1, 85),
    -> (102, 'Karthik', 2, 78),
    -> (103, 'Rahul', 1, 92),
    -> (104, 'Vijay', 3, 88),
    -> (105, 'Ajay', 2, 76);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql>
mysql> SELECT student.student_id,
->         student.student_name,
->         department.dept_name,
->         student.marks
-> FROM student, department
-> WHERE student.dept_id = department.dept_id;
```

student_id	student_name	dept_name	marks
103	Rahul	CSE	92
101	Arun	CSE	85
105	Ajay	ECE	76
102	Karthik	ECE	78
104	Vijay	IT	88

5 rows in set (0.00 sec)

```
mysql>
mysql> SELECT student.student_id,
->         student.student_name,
->         department.dept_name,
->         student.marks
-> FROM student
-> INNER JOIN department
-> ON student.dept_id = department.dept_id;
```

student_id	student_name	dept_name	marks
103	Rahul	CSE	92
101	Arun	CSE	85
105	Ajay	ECE	76
102	Karthik	ECE	78
104	Vijay	IT	88

5 rows in set (0.00 sec)

```
mysql>
mysql> SELECT student.student_id,
->         student.student_name,
->         department.dept_name,
->         student.marks
-> FROM student
-> LEFT OUTER JOIN department
-> ON student.dept_id = department.dept_id;
```

student_id	student_name	dept_name	marks
101	Arun	CSE	85
102	Karthik	ECE	78
103	Rahul	CSE	92
104	Vijay	IT	88
105	Ajay	ECE	76

5 rows in set (0.00 sec)

```
mysql>
mysql> SELECT student.student_id,
->         student.student_name,
->         department.dept_name,
->         student.marks
-> FROM student
-> RIGHT OUTER JOIN department
-> ON student.dept_id = department.dept_id;
```

student_id	student_name	dept_name	marks
103	Rahul	CSE	92
101	Arun	CSE	85
105	Ajay	ECE	76
102	Karthik	ECE	78
104	Vijay	IT	88
NULL	NULL	MECH	NULL

6 rows in set (0.00 sec)